FOOD DONATION BLOCKCHAIN

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Use Case Report

submitted by

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CERTIFICATE

This is to certify that the Use Case report entitled **"Food Donation Blockchain"** is being submitted by **Arepalli Charitha Sri (23501A0512)** as part of **Assignment-2** for the **Blockchain Technology (23CS4701)** course in **4-1** during the academic year **2026-2027**.

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1. INTRODUCTION

1.1. Overview of Blockchain Technology

Blockchain technology is a decentralized and distributed ledger system that records transactions across multiple nodes, ensuring transparency, security, and immutability. Unlike traditional databases, blockchain operates on a peer-to-peer network, eliminating the need for central authorities. Each block in the chain contains a cryptographic hash of the previous block, making data tampering virtually impossible. Originally designed for cryptocurrency transactions, blockchain has now expanded into various domains, including finance, supply chain, healthcare, and digital warranty management.

1.2. Relevance of Blockchain in Food Donation

Blockchain is a distributed ledger technology that records transactions in a verifiable sequence. Once a transaction is confirmed on a blockchain, modifying its recorded history is generally difficult. This property makes blockchain useful in applications where multiple participants need to examine the same transaction records.

In food donation management, blockchain can support traceability by recording details such as the food item, quantity, expiry date, pickup location, and current donation status. Authorized participants can submit updates, and the resulting transactions provide a history of changes. This can reduce dependence on a single editable record maintained by one participant.
Smart contracts are programs deployed on a blockchain that execute predefined rules when users call their functions. In this project, the Solidity smart contract supports donation registration, NGO authorization, and status updates. The prototype uses statuses representing registration, receipt by an NGO, transportation, and distribution.

1.3 Purpose of the Project

The main purpose of Food Donation Blockchain is to demonstrate how blockchain-based records can improve the organization and traceability of food donations. The project combines a smart contract, a local blockchain development environment, a wallet connection, and a browser-based user interface to demonstrate a practical application of blockchain concepts.

The system is an academic prototype. It does not independently verify food quality, guarantee safe transportation, or confirm that every recorded distribution physically occurred. Such verification would require additional operational procedures and evidence from participating organizations.

2. BACKGROUND

Food donation activities require coordination between food providers, collection teams, charitable organizations, and beneficiaries. The usefulness of a donation depends not only on the availability of food but also on timely communication, proper handling, transportation, and distribution. When information is recorded manually, participants may not have a common, reliable view of the donation process.

2.1. Dependence on Manual Records

Donation activities often rely on phone calls, messaging applications, registers, and spreadsheets. Records may be incomplete or scattered across different locations, making information sharing and verification difficult.

2.2. Lack of Transparent Donation Tracking

Participants may not know whether food has been collected, transported, or distributed. A common tracking mechanism can reduce communication delays and provide clear status updates.

2.3. Difficulty in Maintaining Accurate Quantities

Donation records include food types and quantities. When updates are maintained in different systems, discrepancies may occur between the quantity offered, collected, and distributed. Structured records help participants review donation details and track their status.

2.4. Food Expiry and Time Sensitivity

Perishable food must be collected and distributed within an appropriate period. Missing expiry information or delayed communication can affect coordination. Recording expiry dates helps identify time-sensitive donations. However, physical inspection and food-safety procedures remain necessary to determine whether food is safe to consume.

2.5. Accountability Between Participants

Donors and receiving organizations need clear records of donation-handling activities. Manual processes may make it difficult to identify who submitted an update and when. Blockchain transactions provide a verifiable history associated with the submitting wallet address.

2.6. Need for a Digital Solution

A digital donation platform can organize donation details, while blockchain provides a verifiable record of important actions. The Food Donation Blockchain project uses a Solidity smart contract and a React-based interface to make donation registration and status tracking more transparent and auditable.

2.7. Proposed Food Donation Blockchain System

The system uses blockchain and a web interface to manage food donations. The Solidity smart contract stores donation details, assigns unique IDs, and allows authorized NGOs to update donation statuses. React, MetaMask, Ethers.js, and Hardhat support the interface, wallet connection, blockchain interaction, and testing.

2.8. Major Participants

Donors: Provide food details, quantity, expiry date, and pickup location.
NGOs: Collect and distribute food and update donation statuses.
System Administrator: Authorizes NGO accounts and performs administrative operations.
Beneficiaries: Receive donated food but are not individually verified by the prototype.
Application Users: View donation records and perform permitted blockchain transactions.

2.9. Need for Access Control

Donation records should not be updated arbitrarily by any connected account. For example, the ability to authorize an NGO should be restricted to an administrative account, while status updates should follow the contract's authorization rules. Smart-contract access control helps enforce these restrictions at the blockchain level.

2.10. Scope and Limitations

The current prototype demonstrates digital donation registration and status tracking using a local blockchain. It is not yet a complete production platform with verified NGO identities, GPS-based collection tracking, food-quality inspection, automated notifications, or a real-world beneficiary verification mechanism. These functions can be considered for future development.

The proposed solution therefore serves as a proof of concept for applying blockchain technology to food donation management.

3. BLOCKCHAIN BASICS

Blockchain is a distributed ledger technology that records transactions in a sequence of cryptographically linked blocks. Instead of relying exclusively on a single conventional database, a blockchain network maintains records according to the rules of its protocol. Transactions are submitted to the network and included in blocks after being processed according to the network's validation rules.

3.1. Decentralization

Decentralization refers to distributing control and record maintenance across multiple network participants rather than relying entirely on one central authority. Depending on the blockchain network, different nodes may store copies of the ledger and validate transactions.

In a food donation application, decentralization can be useful when donors, NGOs, and other stakeholders need to verify recorded activities. It reduces the need to treat one organization's private database as the only source of transaction history. However, the current project is tested on a local Hardhat network, which is a development environment rather than a live decentralized production network.

3.2. Immutability

Immutability describes the difficulty of altering confirmed blockchain records without following the network's accepted mechanisms. Blockchain systems use cryptographic structures and consensus rules to protect the integrity of transaction history.

For Food Donation Blockchain, donation registration and subsequent status updates are submitted as blockchain transactions. These transactions can be inspected through the development environment, providing a traceable history of the operations performed. Immutability does not guarantee that the information initially entered was correct. If a user enters an incorrect quantity or location, the application still needs appropriate validation and correction procedures.

3.3. Transparency

Transparency means that authorized observers can examine the information made available through the system. On a public blockchain, many transaction details may be publicly visible. In a local development environment, visibility depends on the tools and access available to the participants.

The project uses the smart contract to maintain structured donation records. This supports consistent access to recorded information and makes it easier to demonstrate the sequence of donation operations.

3.4 Traceability

Traceability is the ability to follow a record through different stages of a process. In food donation, a record may move from registration to collection, transportation, and distribution. Recording these stages helps participants understand the progress of each donation.

The prototype models this progression through donation-status updates. Each donation is associated with an identifier that helps distinguish it from other records.

3.5 Smart Contracts

A smart contract is a program deployed on a blockchain. It defines data structures, functions, and rules that determine which operations can be performed and under what conditions. Users interact with the contract by submitting transactions that call its functions.

The Food Donation Blockchain project uses Solidity to implement donation-management logic. The contract stores donation details and supports operations such as registering donations, authorizing NGO addresses, and updating donation status. The contract applies its programmed rules to determine whether a requested operation is permitted.

Smart contracts help reduce dependence on manual enforcement of application rules. Nevertheless, they must be carefully designed and tested because programming errors can affect the system's behavior.

3.6 Transactions and Wallets

A blockchain transaction is an operation submitted by an account. Transactions that change blockchain state generally require a signature from the account authorizing the operation. MetaMask allows users to connect a wallet and approve transactions from a browser-based application.

In this project, MetaMask connects the React frontend to the local blockchain. Ethers.js provides the interface used by the application to call contract functions and read blockchain data. The connected account determines which wallet address submits a transaction, and the smart contract checks the permissions associated with that caller.

3.7 Cryptographic Security

Blockchain networks use cryptographic techniques to protect transaction integrity and authenticate transactions. Wallet addresses are associated with cryptographic keys, and a transaction signature demonstrates authorization by the corresponding account.

The application should never request a user's private key directly or store private keys in its frontend code. Development accounts and their test keys must be kept separate from accounts containing real assets. Local test accounts are appropriate for demonstrations but must not be reused for production transactions.

4. USE CASE OVERVIEW

The use case for a Blockchain-Based Food Donation Tracking System aims to improve the management and traceability of food donations by leveraging blockchain technology. The system provides a structured platform for registering food donations, maintaining donation details, authorizing non-governmental organizations (NGOs), and tracking the progress of food distribution. By using smart contracts, the system supports transparent record keeping, controlled access, and verifiable donation-status updates.

4.1. Objectives

The primary objective of this blockchain-based food donation tracking system is to provide a digital platform for registering and monitoring food donations. Traditional methods rely on paper records, spreadsheets, phone calls, and messaging applications, making it difficult to maintain accurate donation details. Blockchain technology enables donation information to be recorded through a smart contract and accessed via a web application.

A key challenge in food donation management is the lack of transparency in tracking food from donors to receiving organizations. The proposed system maintains donation records with unique identifiers and status information. Blockchain transactions provide a verifiable history of recorded operations, helping participants monitor donation progress and reducing dependence on manual records.

Another important objective is to improve accountability through NGO authorization. The smart contract allows the administrator to authorize NGO wallet addresses and restrict operations according to predefined rules. This ensures that only permitted participants can perform specific status updates, reducing unauthorized changes.

The system also aims to improve donation tracking through stages such as Registered, Received by NGO, In Transit, and Distributed. These stages help users understand the latest recorded status and monitor progress throughout the distribution process. Unique donation identifiers make records easier to reference and verify, improving coordination between donors and charitable organizations.

Transparency and reliable record keeping are additional objectives. The smart contract stores donation information, while the web interface enables convenient interaction. By combining Solidity, Hardhat, MetaMask, Ethers.js, and React, the application demonstrates how blockchain can support better coordination and auditing of food donations. However, physical collection, food safety, and actual delivery still require real-world verification.

4.2. Scope of the System

The blockchain-based food donation tracking system focuses on registering and tracking food donations through a web application connected to a smart contract. The system includes:

1. **Food Donors:** Provide food name, quantity, expiry date, and pickup location..
2. **NGOs and Charitable Organizations:** Coordinate food collection and update donation status through authorized accounts.
3. **Contract Owner / Administrator:** Authorizes NGO wallet addresses and performs permitted administrative operations.
4. **Blockchain Network:** Stores donation records and transaction history for traceability.
5. **Web Application:** Enables wallet connection, donation registration, record viewing, and status tracking.

4.3. System Architecture

The architecture of the Blockchain-Based Food Donation Tracking is illustrated in **Figure 4.1** and consists of the following components:

4.3.1. User Layer

This layer includes the participants who interact with the web application. (**Figure 4.1 - User Layer**).

- **Food Donors:** Register donations and review their recorded details.
- **NGOs and Charitable Organizations:** View donation records and update status through authorized accounts.
- **Contract Owner / Administrator:** Authorizes NGO wallets and performs permitted administrative operations.

4.3.2. Web Application Layer

This layer acts as the interface between users and the blockchain. (**Figure 4.1 - Web Application Layer**)

- **User Interface (Frontend):** Displays donation details, quantities, expiry dates, pickup locations, and statuses
- **Donation Management System:** Collects donation information, retrieves records, and presents data using React and Ethers.js.
- **Wallet Integration:** Uses MetaMask to connect wallets and request transaction approval for supported operations.

4.3.3. Blockchain Layer

This layer ensures security, transparency, and immutability of warranty transactions. (**Figure 4.1 - Blockchain Layer**).

- **Smart Contract Layer**:
  - **FoodDonation Contract**:
    - Stores donation ID, food name, quantity, expiry date, pickup location, and status
    - Provides functions for donation registration, NGO authorization, and status updates.
  - **Smart Contract Execution**:
    - Creates donation records with unique identifiers upon registration..
    - Records status changes made by authorized NGOs.
    - Restricts operations according to contract permissions.
- **Blockchain Network**:
  - **Transaction Records**: Maintains donation registration and status-update transactions.
  - **Traceable Updates:** Allows donation progress to be reviewed using its ID and contract data.
  - **Local Development Network:** Uses Hardhat for testing, with MetaMask and Ethers.js enabling frontend interaction.

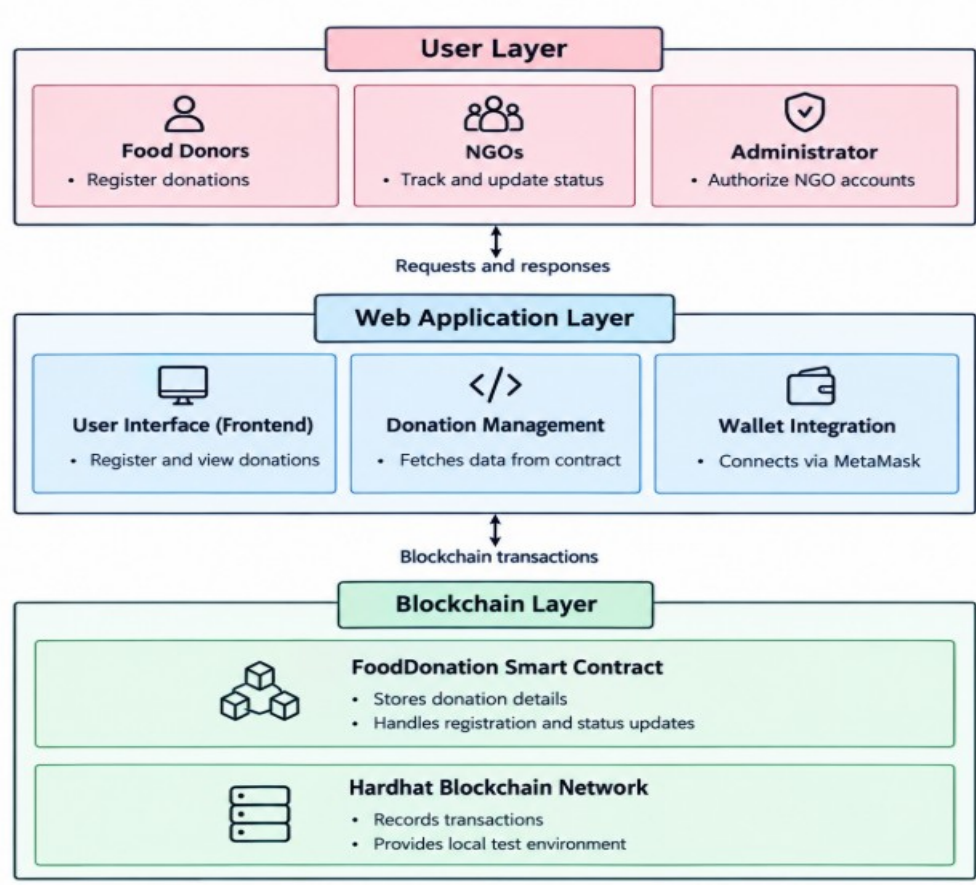


Figure 4.1: Architecture of Blockchain based Food Donation Tracking System

5.IMPLEMENTATION

5.1. Setting Up the Blockchain Environment

The food donation tracking system uses Ethereum-compatible blockchain technology to maintain transparent donation records. Solidity is used to develop smart contracts, while Hardhat provides the environment for compiling, deploying, and testing them on a local blockchain network.

Node.js and the required libraries support development, while React.js provides the user interface. MetaMask connects users' wallets and approves transactions, and Ethers.js enables the frontend to interact with the smart contract.

After deployment, the contract address and Application Binary Interface (ABI) are integrated into the frontend. This setup allows donation registration, record retrieval, and status updates to be tested locally without using a public blockchain.

5.2. Defining Smart Contracts

5.2.1. Writing the Smart Contract for Food Donation Tracking

The FoodDonation smart contract is the core component of the system. It stores donation details, assigns unique identifiers, authorizes NGO accounts, and manages donation status updates.

Each donation contains the donor address, food name, quantity, expiry date, pickup location, and current status. The status progresses through Registered, Received by NGO, In Transit, and Distributed, providing a traceable record of donation activities.

The following Solidity structure represents a donation record:

```
struct Donation {
    uint256 id;
    address donor;
    string foodName;
    uint256 quantity;
    string expiryDate;
    string
        pickupLocation;
    uint256 status;
}
```

A mapping stores donation records using their unique identifiers:

```
mapping(uint256 => Donation) public donations;
```

5.2.2. Registering Participants and Donations

NGOs collect and distribute donated food. The smart contract includes an authorization mechanism to restrict donation-status updates to permitted accounts.

The contract owner can authorize NGO wallet addresses. Authorized accounts can perform permitted operations, while unauthorized accounts are prevented from modifying restricted records. Blockchain transactions also provide a history of submitted updates. However, wallet authorization alone does not verify an NGO's real-world identity.

An illustrative authorization mapping is:

```
mapping(address => bool) public authorizedNGOs;
```

The contract checks authorization before allowing restricted operations:

```solidity
require(
    authorizedNGOs[msg.sender],
    "NGO not authorized"
);
```

5.2.3. Issuing Warranty Tokens

Once The donation registration function allows donors to submit food details through the frontend. The donor enters the food name, quantity, expiry date, and pickup location, then approves the transaction using MetaMask.

The smart contract assigns a unique donation ID and stores the submitted information with an initial status of Registered. Users can retrieve and track the donation through the application. These records help NGOs coordinate collection and prioritize time-sensitive donations. However, the accuracy of the information depends on user input, and blockchain cannot independently verify food quality or physical quantity.

An illustrative function signature is:

```
function registerDonation(
    string memory foodName,
    uint256 quantity,
    string memory expiryDate,
    string memory pickupLocation
) external;
```

5.2.4. Viewing Registered Donations

The system allows users to retrieve registered donations using their unique identifiers. The frontend displays details such as donation ID, food name, quantity, expiry date, pickup location, donor address, and current status.

React communicates with the smart contract through Ethers.js to retrieve and display donation records. Read operations do not require transaction approval, while blockchain updates require authorization through the connected wallet.

The retrieval process involves the following steps:

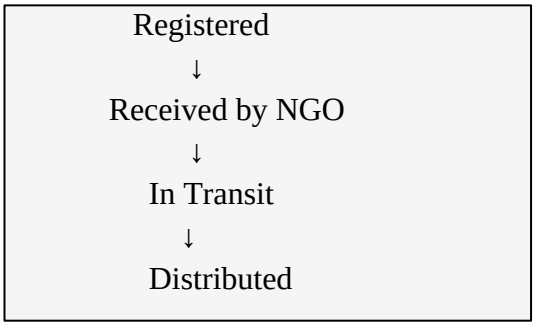
1. Obtain the donation identifier.
2. Connect to the deployed FoodDonation contract.
3. Request the donation record from the contract.
4. Read the returned donation details.
5. Display the details and current status in the frontend.

5.2.5. Updating the Donation Status

The system tracks donation progress through predefined statuses. Authorized participants can update the status according to the smart contract's permissions.

- **Status 0 – Registered:** Donation details are recorded.
- **Status 1 – Received by NGO:** Receipt is recorded by the NGO.
- **Status 2 – In Transit:** The donation is marked as being transported.
- **Status 3 – Distributed:** The donation is marked as distributed.

The status flow is represented as follows:



When an authorized user submits an update, the smart contract verifies permissions and records the change. The recorded status helps track donation progress but does not independently verify physical collection, transportation, or delivery.

5.2.6. Tracking Donations Throughout the Distribution Process

A key feature of the proposed system is the ability to track a donation from registration through the distribution stage. In a conventional donation process, information may be maintained in separate registers, spreadsheets, or messages. This can make it difficult to obtain a consistent view of a donation's progress.

The blockchain-based approach stores the donation record and its status updates through a smart contract. The unique donation identifier helps participants refer to the same record during collection and distribution. As authorized users submit updates, the latest status can be retrieved and displayed through the frontend.

For example, when a donor registers a quantity of food, the system initially records the donation as Registered. After collection, an authorized participant can update it to Received by NGO. During transportation, the status can be changed to In Transit. When the responsible participant records completion, the donation can be marked as Distributed.

These updates support transparency by making the recorded progress easier to review. Since transactions are recorded on the blockchain, previous confirmed transactions can be inspected rather than relying only on a single editable status displayed by a centralized application.

The prototype demonstrates this sequence using a local Hardhat blockchain. In a real deployment, additional features such as participant verification, transaction monitoring, event logging, and controls for correcting inaccurate reports would be required to strengthen the reliability of the process.

5.3. Deploying and Integrating with a Frontend

The FoodDonation smart contract is compiled and deployed using Hardhat on a local blockchain network. The deployment generates a contract address, which is configured in the React frontend along with the Application Binary Interface (ABI).

React provides forms for registering donations and viewing their details and status. Ethers.js connects the frontend to the smart contract, while MetaMask connects user wallets and authorizes transactions.When a user registers a donation or updates its status, the frontend submits the transaction and displays the result after confirmation. Correct contract configuration ensures successful communication between the frontend and blockchain.

The prototype is designed for local testing. Public deployment would require additional security testing, network configuration, and access-control validation.

5.4. Workflow:

The workflow of the Blockchain-Based Food Donation Tracking System, illustrated in Figure 5.1, describes the interactions between donors, NGOs, and the blockchain application. It begins when a donor enters available food details through the frontend, which submits the information to the FoodDonation smart contract.

The smart contract stores the donation details and assigns a unique identifier. Users can use this ID to retrieve the donation record and view its current status through the frontend.

When an NGO receives the food, an authorized participant updates the status. The donation progresses through Received by NGO, In Transit, and Distributed, according to the updates recorded in the system. The smart contract enforces the implemented access-control rules.

MetaMask approves transactions, while Ethers.js connects the React frontend to the smart contract. Invalid transactions are rejected, and successful updates can be retrieved and displayed through the application.

This workflow improves donation traceability and record keeping. However, physical delivery and food safety still require real-world verification.

Figure 5.1: Workflow of Blockchain-Based Food Donation Tracking System

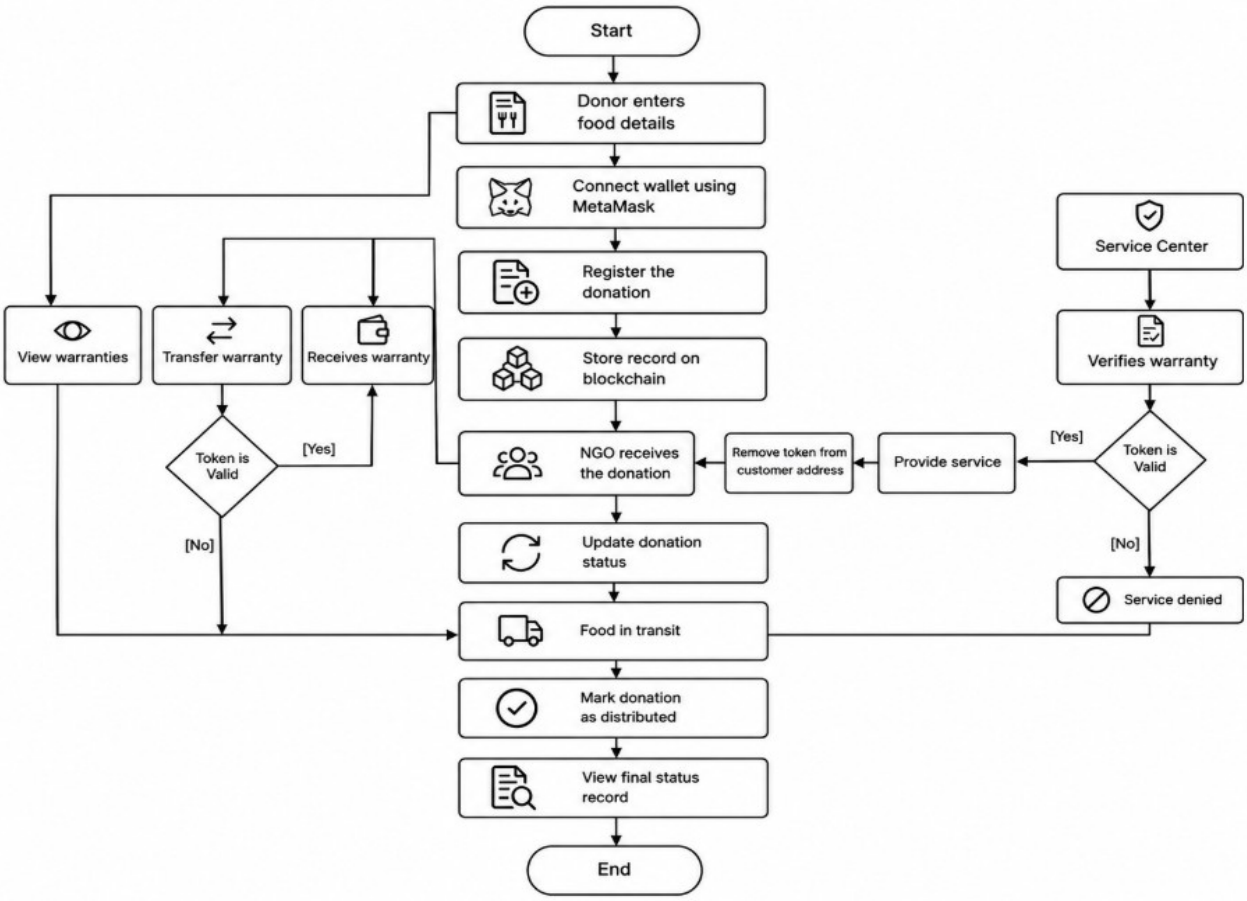


Figure 5.1: Workflow of Blockchain-Based Food Donation Tracking System

6. BENEFITS

Blockchain technology improves transparency and traceability in food donation management. Unlike traditional methods that rely on paper records and spreadsheets, the proposed system stores donation details and status updates through a smart contract, providing a structured digital record for authorized participants.The main benefits of the Blockchain-Based Food Donation Tracking System are described below.

6.1. Enhanced Security and Immutability

Blockchain maintains a transaction history that is difficult to alter without leaving evidence. Donation registrations and status updates are recorded on the blockchain, reducing unauthorized changes to confirmed records. Smart contracts enforce access-control rules, although blockchain cannot guarantee the accuracy of information entered by users.

6.2. Improved Transparency and Traceability

Donation records can be retrieved using unique identifiers, allowing participants to view details such as food name, quantity, pickup location, expiry date, and current status. This helps track donations from registration to distribution, reduces discrepancies between separate records, and makes recorded progress easier to verify.

6.3. Better Coordination Between Donors and NGOs

The system provides a common platform for recording donations and distribution updates. Donors register available food, while authorized NGOs update collection progress. Accurate details about quantity, pickup location, and expiry dates help participants plan collections and reduce repeated communication.

6.4. Reduced Paperwork and Manual Record-Keeping

The digital platform reduces dependence on physical registers, printed forms, and spreadsheets. Users can register and view donations through the web interface, while blockchain maintains transaction records. This simplifies information retrieval and improves donation management

6.5. Improved Accountability

Blockchain transactions are associated with the addresses that submit them, helping identify which accounts register donations or update statuses. This transaction history supports accountability and helps review discrepancies, although wallet addresses alone do not verify real-world identities..

6.6. Faster Access to Donation Information

The application allows users to retrieve donation records using unique identifiers instead of searching through manual documents. Participants can quickly check quantities, pickup locations, expiry dates, and current statuses, provided the application and blockchain network are available.

6.7. Support for Efficient Food Distribution

Recording food quantities and expiry dates helps NGOs identify donations that may require prompt collection and distribution. The information supports planning and coordination, while physical inspection, proper storage, and food-safety procedures remain essential.

6.8. Reduced Risk of Undetected Record Changes

Blockchain maintains a transaction history that makes retrospective changes more difficult to conceal. Combined with smart-contract validation and access controls, it helps protect donation records against unauthorized updates. Security also depends on correct contract implementation and wallet management.

6.9. Increased Trust Among Participants

A shared, traceable record allows donors and NGOs to review donation details and recorded progress. This reduces confusion and supports better coordination. Reliable data entry, participant verification, and confirmation of actual deliveries are still necessary to maintain trust.

6.10. Support for Social Responsibility and Sustainability

The system helps organize donation records and connect available food with intended recipients. By improving coordination and reducing administrative effort, it may support food donation activities and waste reduction. Its actual impact depends on real-world implementation and measurable outcomes.

Overall, the blockchain-based food donation tracking system combines smart contracts, a web interface, and wallet-based transaction approval to create a traceable record of donation activity. The prototype demonstrates the technical workflow, while larger-scale deployment would require further testing and operational improvements.

7. CHALLENGES

Although blockchain-based food donation tracking improves transparency and record management, challenges remain in scalability, data accuracy, user adoption, internet connectivity, and food safety. Below are some of the key challenges:

7.1. Scalability Issues

High transaction volumes may increase processing time and costs. The local Hardhat network supports development and testing but does not demonstrate large-scale performance. Future versions can explore scalable networks, transaction batching, and efficient smart contract design.

7.2. Data Accuracy and Verification

Blockchain preserves submitted records but cannot verify their accuracy. Incorrect donation details may remain in the transaction history, and a Distributed status does not prove actual delivery. Verified accounts, digital acknowledgements, and confirmation from receiving organizations can improve reliability.

7.3. Lack of User Awareness and Adoption

Some donors and NGO workers may find blockchain, digital wallets, and transaction approvals unfamiliar. A simple interface, clear instructions, and basic training can make the system easier to use.

7.4. Dependence on Internet Connectivity

The application requires internet access to interact with the blockchain. Poor connectivity may delay donation registration and status updates. Future versions can explore offline entry and synchronization while clearly distinguishing pending changes from confirmed transactions.

7.5. Food Safety and Expiry Management

Recording expiry dates does not guarantee food safety. Storage, packaging, temperature, and handling also affect food quality. Real-world implementation requires appropriate inspections and food-safety procedures.

Despite these challenges, blockchain provides a useful foundation for traceable donation records. Addressing scalability, data verification, user adoption, connectivity, and food safety is essential for reliable real-world use.

8. CONCLUSION

The Blockchain-Based Food Donation Tracking System aims to improve the recording, transparency, and traceability of food donations. Traditional donation management often relies on manual records and separate communication channels, making it difficult to track donation details and distribution progress. The proposed system uses the FoodDonation smart contract to register donations, assign unique identifiers, and record status updates.

The prototype integrates Solidity, Hardhat, React.js, Ethers.js, and MetaMask to provide a web-based donation tracking interface. It supports four stages: Registered, Received by NGO, In Transit, and Distributed. These features help improve information sharing and coordination between donors and NGOs. However, blockchain cannot independently verify physical delivery or food quality. Security testing, participant verification, food-safety procedures, and scalability evaluation are necessary before real-world deployment.

8.1. Future Outlook for Enhancements

The system can be enhanced by introducing stronger identity verification for donors and NGOs. Verified organizational accounts could help associate blockchain addresses with recognized participants and reduce the risk of unauthorized access.

Another possible enhancement is the integration of QR codes for donation identification. A QR code could represent a donation identifier, allowing authorized participants to retrieve the corresponding record more conveniently during collection or distribution. Appropriate authentication would still be needed to prevent unauthorized status updates.

The platform could also include notifications for approaching expiry dates, pending collections, and delayed status updates. These features could help participants prioritize time-sensitive donations and coordinate collection schedules more effectively.

In future versions, IoT sensors could be evaluated for monitoring selected storage or transportation conditions where suitable hardware is available. Such information would require careful integration and validation before being treated as evidence of food safety.

Finally, the prototype could be evaluated on an appropriate test network and later adapted for a production environment after security audits, performance testing, and operational review. These enhancements could help evolve the project into a more reliable platform for transparent food donation management.

9. SDG's ADDRESSED

The Blockchain-Based Food Donation Tracking System supports selected United Nations Sustainable Development Goals (SDGs) by improving donation records and tracking distribution updates. The project is relevant to reducing hunger, promoting responsible consumption, and encouraging technological innovation.

SDG 2: Zero Hunger

The project supports SDG 2 by providing a structured method to record available food donations and track their reported progress. Details such as food quantity, expiry date, and pickup location help donors and NGOs coordinate collection and distribution. Although the system cannot guarantee delivery to people in need, it can support more organized donation management.

SDG 12: Responsible Consumption and Production

SDG 12 focuses on responsible resource use and reducing waste. The system records donation details and status updates, helping participants monitor distribution progress and prioritize food nearing its expiry date. Its contribution to reducing food waste depends on accurate records and effective collection and distribution.

SDG 9: Industry, Innovation, and Infrastructure

SDG 9 promotes innovation and sustainable technological development. This project integrates Solidity, Hardhat, React.js, Ethers.js, and MetaMask to demonstrate how blockchain and smart contracts can support transparent donation record management. Further development and testing could improve the system's reliability and practical application.

Overall, the project demonstrates how digital technologies can support social responsibility, efficient resource management, and innovation. Its actual impact on these SDGs would depend on successful implementation and measurable results.

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11. APPENDIX A

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